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DC Motor No-Start Fault Tree & Test Sheet

For maintenance technicians and engineers troubleshooting PMDC and geared motors that will not start

Complete Step 1 before any test. Then apply the run command, measure motor current and terminal voltage, and follow the branch those two readings indicate. Record every reading.

Step 1 — Make it safe

- [ ] Lock out and tag out the supply; prove dead with a tested meter.
- [ ] Allow controller capacitors to discharge and verify zero volts on the DC bus.
- [ ] Isolate battery packs at the battery terminals or approved disconnect.
- [ ] Support suspended loads mechanically before releasing any brake; release spring energy in gates and barriers.
- [ ] Disconnect the motor from the controller before turning the shaft by hand; a PMDC motor generates voltage when turned.

Step 2 — Branch A: no current

- [ ] Measure supply voltage with the run command on: expect nameplate voltage ± 10 %.
- [ ] Measure voltage at the motor terminals: 0 V means the break is upstream (fuse, breaker, contactor, connector, controller enable).
- [ ] Check controller inputs: interlocks, E-stop, direction, speed reference range (0–5 V or 0–10 V), latched faults, throttle-at-zero inhibit.
- [ ] Full voltage at terminals with 0 A means an open circuit inside the motor.
- [ ] Isolated: measure terminal resistance with a four-wire or zeroed low-resistance meter; open reading → brushes, shunt, protector or armature lead.
- [ ] Inspect brush length, freedom in holders and shunt condition.

Step 3 — Branch B: current flows, shaft does not turn

- [ ] Isolate, uncouple and release the brake; turn the shaft by hand. Tight or rough → bearing, corrosion or gear jam.
- [ ] Brake: confirm rated coil voltage at the brake terminals and an audible release click.
- [ ] Measure motor terminal voltage during the attempted start and compare with supply voltage; the difference is lost in cables and joints.
- [ ] Calculate starting current: $I = (V - \text{brush drop}) / (R_a + R_{cable} + R_{joints})$. Starting torque $T = K_t \times I$.
- [ ] Cable resistance: $R = 0.0172 \times \text{loop length (m)} / \text{area (mm}^2\text{)}$ ohms for copper at 20 °C.
- [ ] Check the controller current limit: starting torque cannot exceed $K_t \times$ current limit.
- [ ] Check no-load speed at rated voltage; clearly above the test-sheet no-load speed → weakened magnets.

Step 4 — Branch C: starts only when pushed or tapped

- [ ] Isolated: turn the shaft slowly through one revolution while reading terminal resistance.
- [ ] Reading doubles or goes open at certain positions → dead spot (open coil, lifted riser or burnt bar).
- [ ] Measure bar-to-bar resistance around the commutator; readings should agree within a few per cent.

- [ ] High bar pair → open coil. Low bar pair → shorted coil.
- [ ] Starts after tapping the frame → brush sticking in holder; clean holder and check brush length.

Step 5 — Branch D and final confirmation

- [ ] Fuse blows instantly: stop re-energising; test insulation resistance winding-to-frame at 500 V DC.
- [ ] After monsoon standstill: measure insulation resistance and dry the winding if low before applying power.
- [ ] Bench test: motor disconnected and uncoupled on a current-limited supply at reduced voltage; smooth running clears the motor.
- [ ] Record the root cause, the corrective action and the readings after repair.

Test, Expected Reading and Conclusion

Test	Expected Reading	If Not, Conclusion
Supply voltage, run command on	Nameplate voltage ± 10 %	Supply, fuse, breaker or battery fault
Motor terminal voltage, run command on	Close to supply voltage	0 V: upstream open. Low: cable or joint drop
Motor current, run command on	Rises, then falls as shaft accelerates	0 A: open circuit. High and steady: locked shaft
Terminal resistance, shaft turned slowly	Close to test sheet; steady	Open: brushes or armature. Doubling: dead spot
Bar-to-bar resistance	Within a few per cent	High pair: open coil. Low pair: shorted coil
Insulation resistance, 500 V DC	High and stable	Low: moisture or earth fault
Shaft by hand, uncoupled, brake released	Smooth and free	Bearing, corrosion or gear jam
Bench supply, uncoupled, reduced voltage	Runs smoothly	Fails: motor fault. Runs: controller, wiring or load
No-load speed at rated voltage	Close to test-sheet no-load speed	Clearly higher: weakened magnets

Readings are typical guidance. Always compare with the motor test sheet and nameplate, and with the controller and brake manufacturer's data. Live measurements must be made only by competent persons using correctly rated instruments.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.